CSA0618-DESIGN AND ANALYSIS OF ALGORITHMS

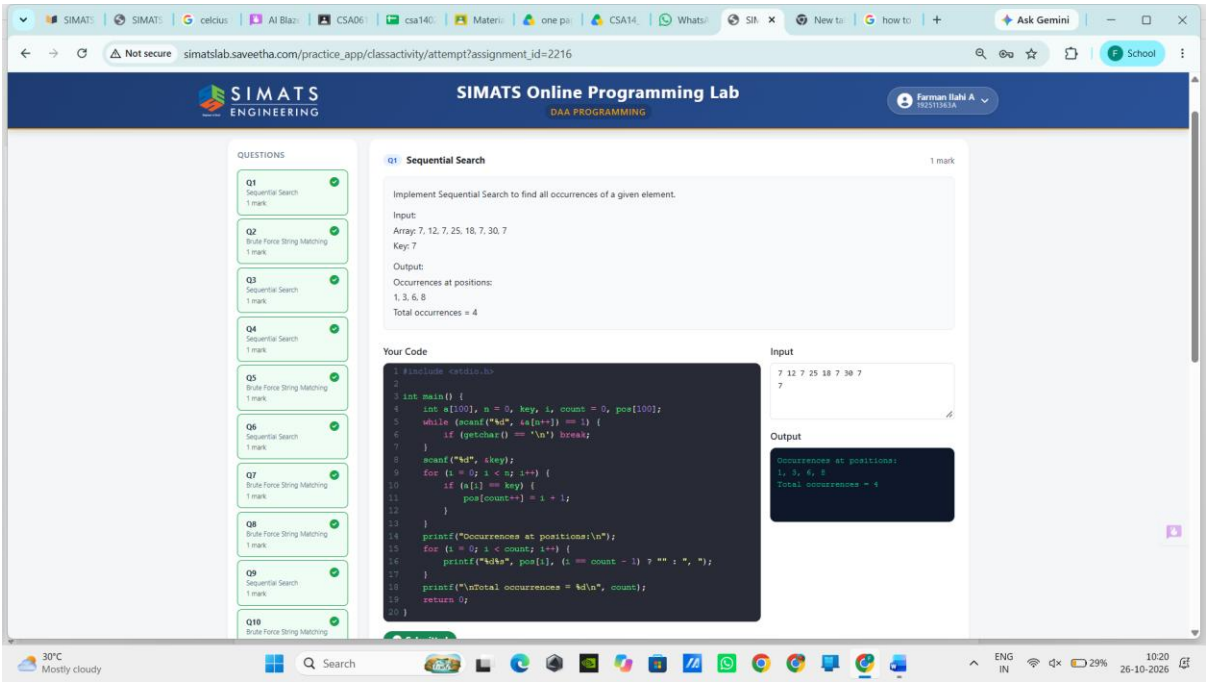
ASSESSMENT OUTPUTS

TOPIC -3

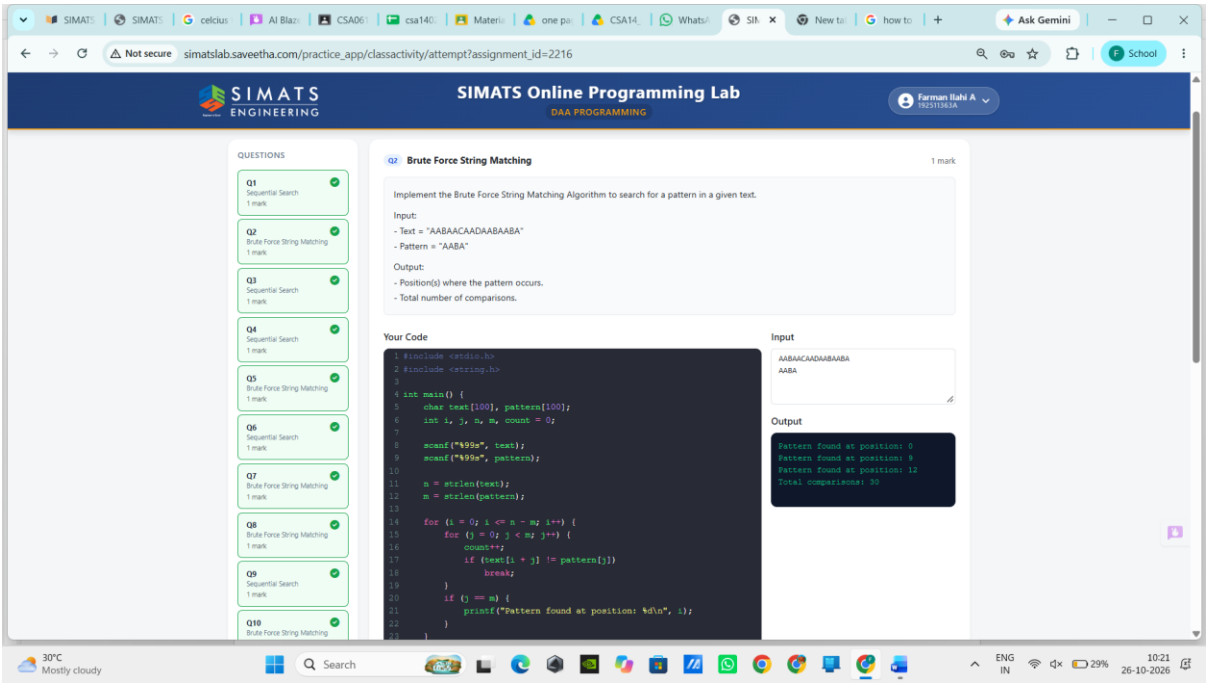
NAME: FARMAN ILAHI A

REG NO: 192511363

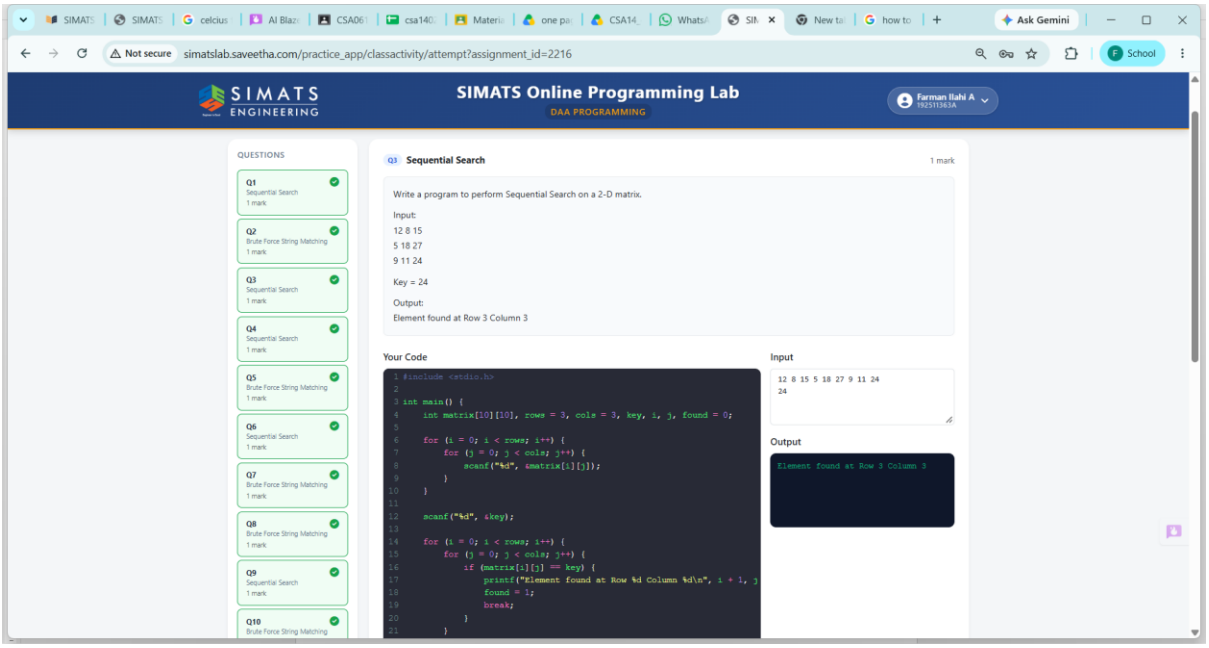
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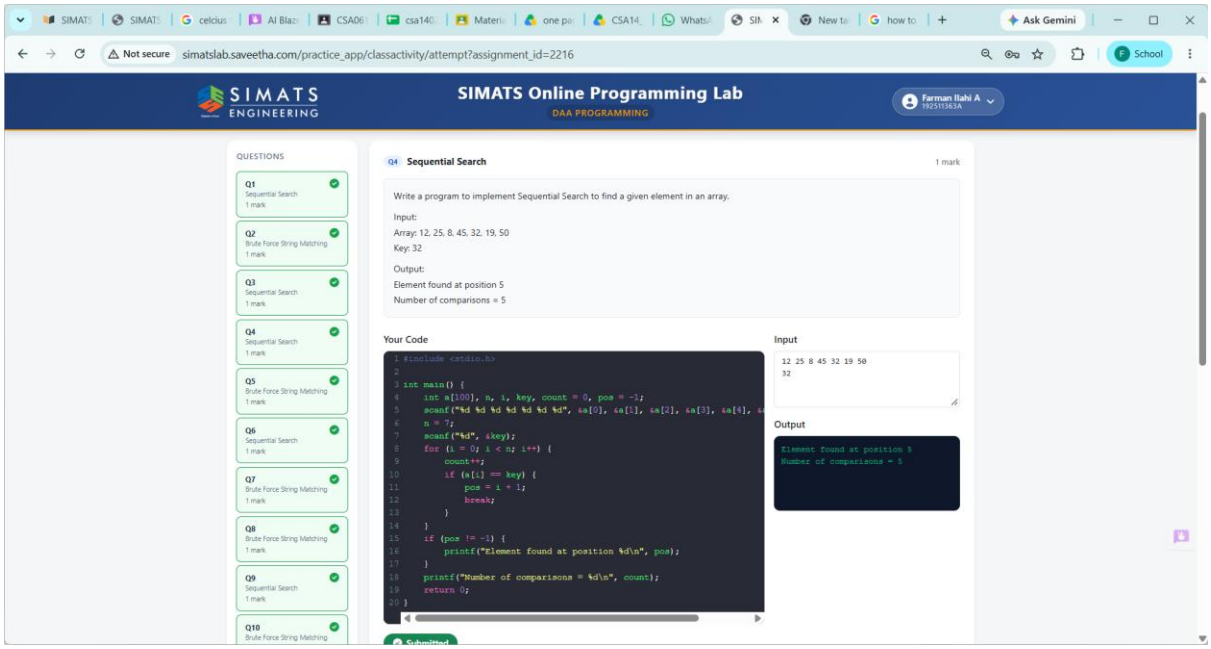
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QUESTIONS
Q2 Brute Force String Matching
1 mark
Implement the Brute Force String Matching Algorithm to search for a pattern in a given text.
Input:
- Text = "AABAACAADAABAABA"
- Pattern = "AABA"
Output:
- Position(s) where the pattern occurs.
- Total number of comparisons.
Your Code
Input
AABAACAADAABAABA
AABA
Output
Pattern found at position: 0
Pattern found at position: 9
Pattern found at position: 12
Total comparisons: 30

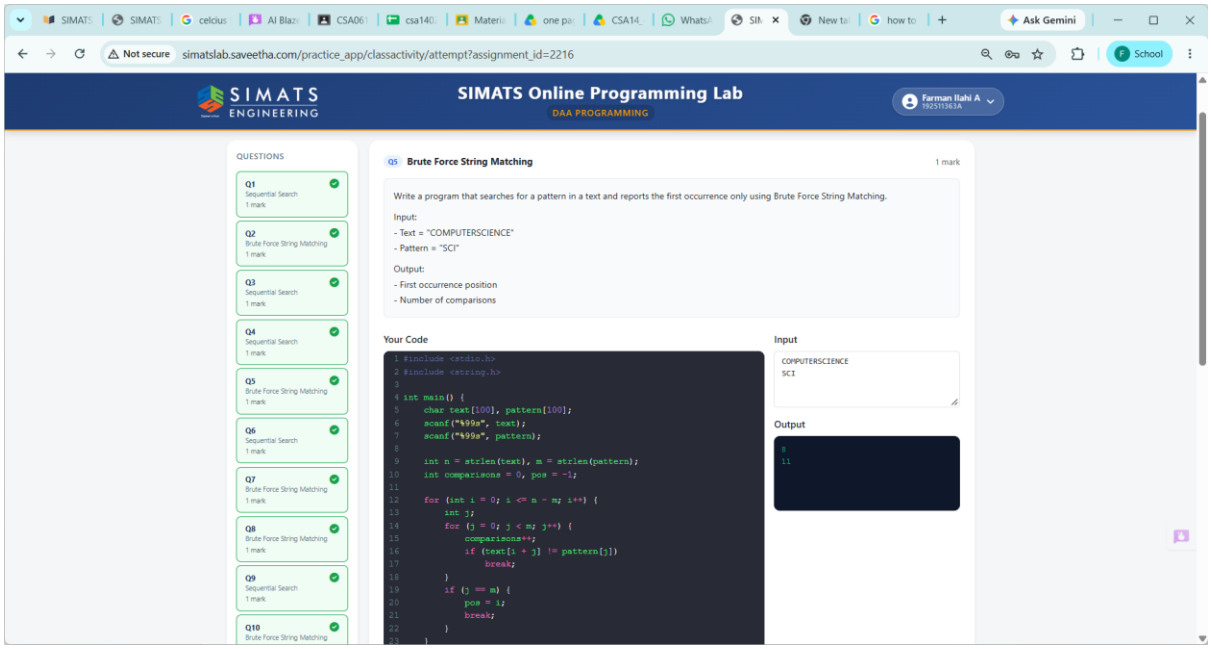
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QUESTIONS
Q3 Sequential Search
1 mark
Write a program to perform Sequential Search on a 2-D matrix.
Input:
12 8 15
5 18 27
9 11 24
Key = 24
Output:
Element found at Row 3 Column 3
Your Code
Input
12 8 15 5 18 27 9 11 24
24
Output
Element found at Row 3 Column 3

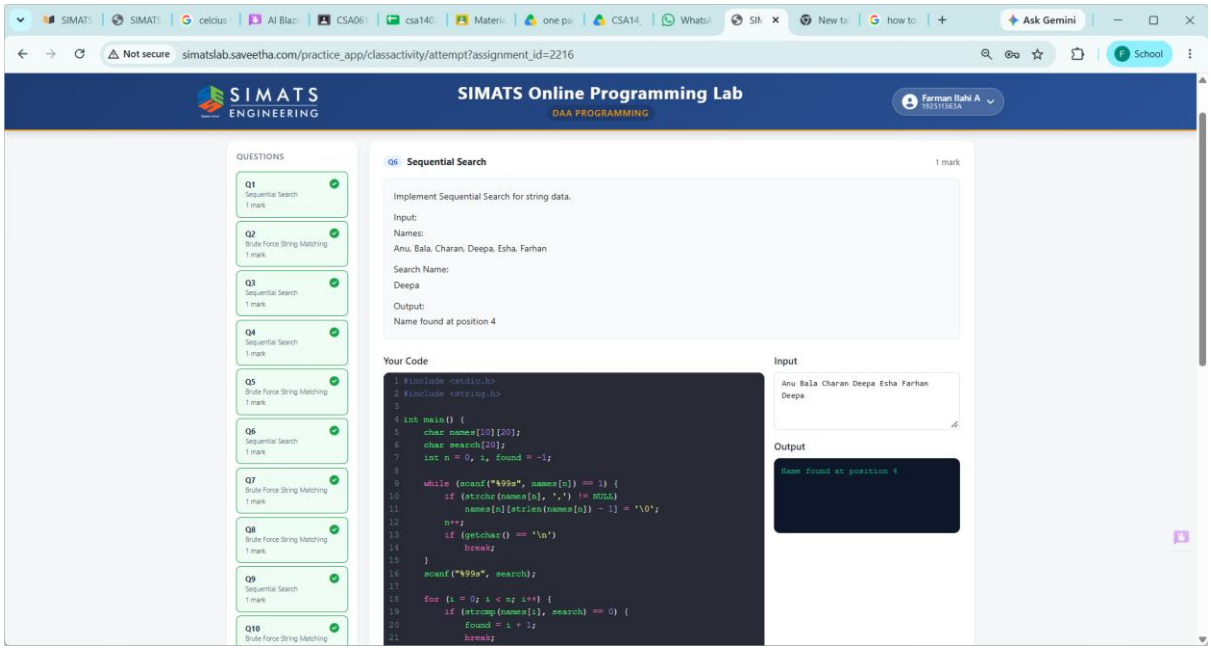
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QUESTIONS
Q4 Sequential Search
1 mark
Write a program to implement Sequential Search to find a given element in an array.
Input:
Array: 12, 25, 8, 45, 32, 19, 50
Key: 32
Output:
Element found at position 5
Number of comparisons = 5
Your Code
Input
12 25 8 45 32 19 50
32
Output
Element found at position 5
Number of comparisons = 5

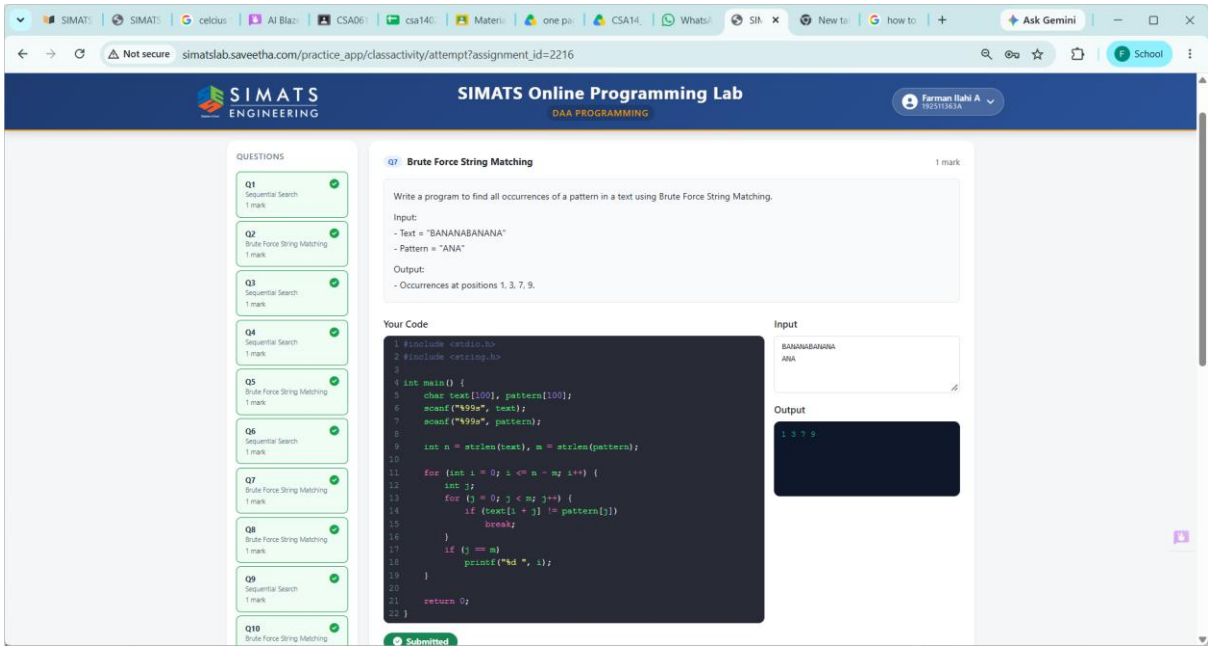
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QUESTIONS
Q5 Brute Force String Matching
1 mark
Write a program that searches for a pattern in a text and reports the first occurrence only using Brute Force String Matching.
Input:
- Text = "COMPUTERSCIENCE"
- Pattern = "SCI"
Output:
- First occurrence position
- Number of comparisons
Your Code
Input
COMPUTERSCIENCE
SCI
Output
8
11

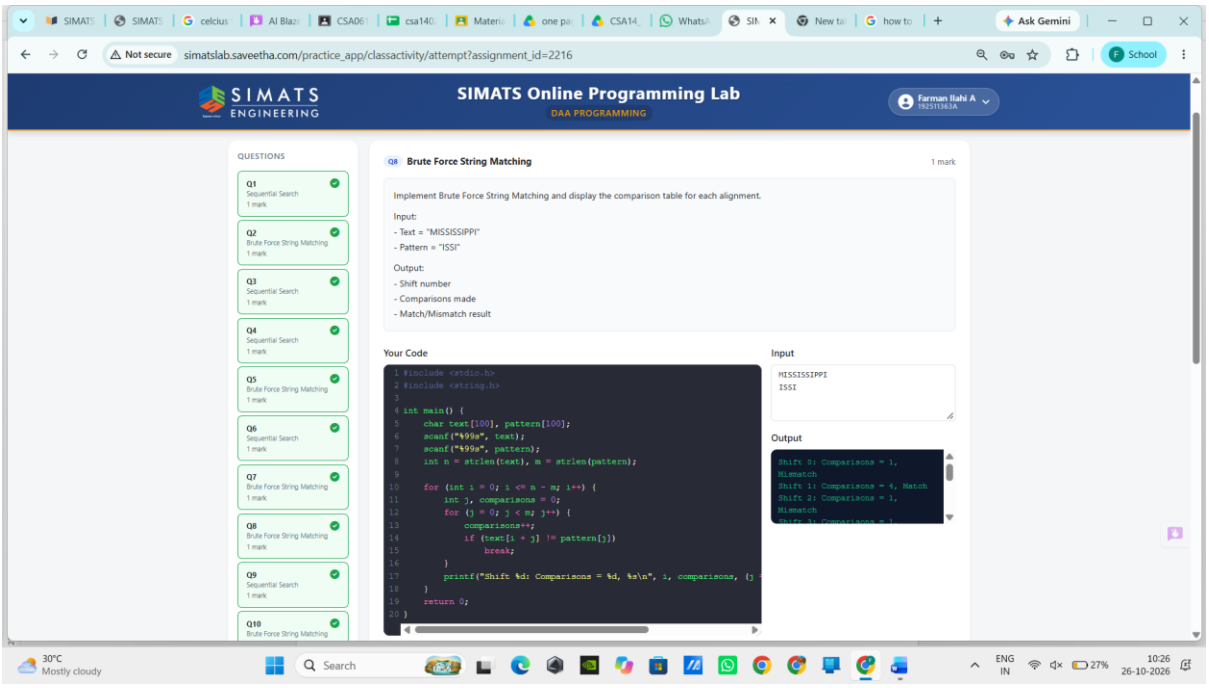
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QUESTIONS
Q1 Sequential Search 1 mark
Q2 Brute Force String Matching 1 mark
Q3 Sequential Search 1 mark
Q4 Sequential Search 1 mark
Q5 Brute Force String Matching 1 mark
Q6 Sequential Search 1 mark
Q7 Brute Force String Matching 1 mark
Q8 Brute Force String Matching 1 mark
Q9 Sequential Search 1 mark
Q10 Brute Force String Matching
Q6 Sequential Search
1 mark
Implement Sequential Search for string data.
Input:
Names:
Anu, Bala, Charan, Deepa, Esha, Farhan
Search Name:
Deepa
Output:
Name found at position 4
Your Code
#include <stdio.h>
#include <string.h>
int main() {
char names[20][20];
char search[20];
int n = 0, i, found = -1;
while (scanf("%99s", names[n]) == 1) {
if (strchr(names[n], ',') != NULL)
names[n][strlen(names[n]) - 1] = '\0';
n++;
if (getchar() == '\n')
break;
}
scanf("%99s", search);
for (i = 0; i < n; i++) {
if (strcmp(names[i], search) == 0) {
found = i + 1;
break;
Input
Anu Bala Charan Deepa Esha Farhan
Deepa
Output
Name found at position 4

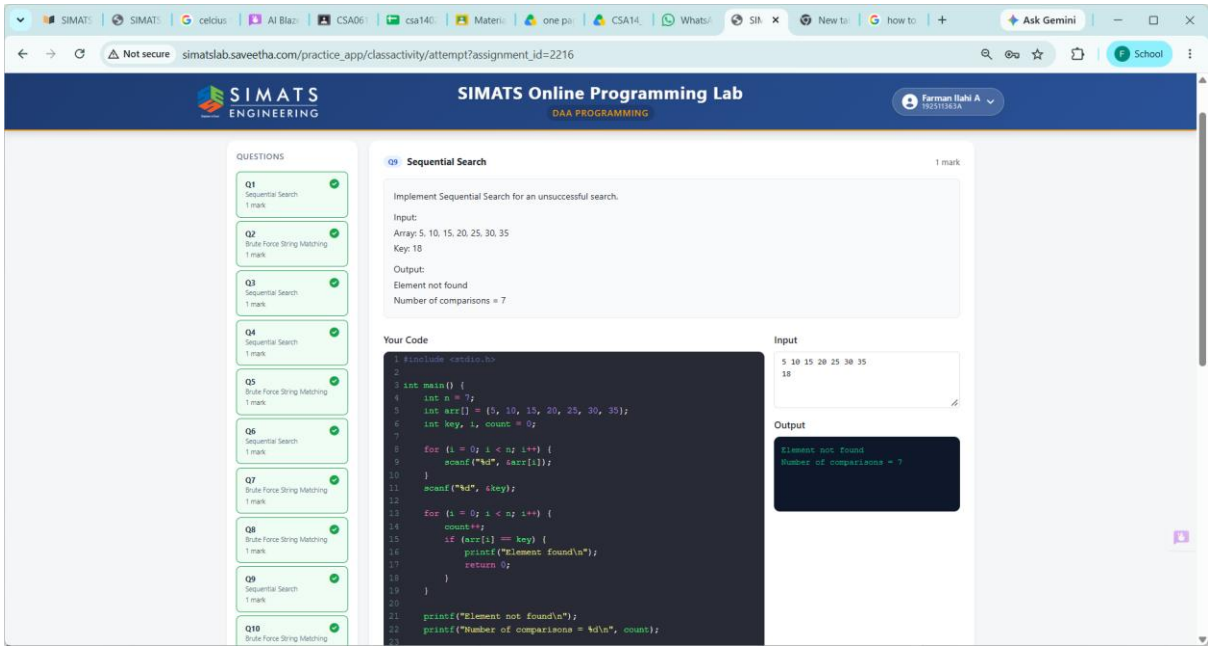
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QUESTIONS
Q1 Sequential Search 1 mark
Q2 Brute Force String Matching 1 mark
Q3 Sequential Search 1 mark
Q4 Sequential Search 1 mark
Q5 Brute Force String Matching 1 mark
Q6 Sequential Search 1 mark
Q7 Brute Force String Matching 1 mark
Q8 Brute Force String Matching 1 mark
Q9 Sequential Search 1 mark
Q10 Brute Force String Matching
Q7 Brute Force String Matching
1 mark
Write a program to find all occurrences of a pattern in a text using Brute Force String Matching.
Input:
- Text = "BANANABANANA"
- Pattern = "ANA"
Output:
- Occurrences at positions 1, 3, 7, 9.
Your Code
#include <stdio.h>
#include <string.h>
int main() {
char text[100], pattern[100];
scanf("%99s", text);
scanf("%99s", pattern);
int n = strlen(text), m = strlen(pattern);
for (int i = 0; i <= n - m; i++) {
int j;
for (j = 0; j < m; j++) {
if (text[i + j] != pattern[j])
break;
}
if (j == m)
printf("%d ", i);
}
return 0;
}
Submitted
Input
BANANABANANA
ANA
Output
1 3 7 9

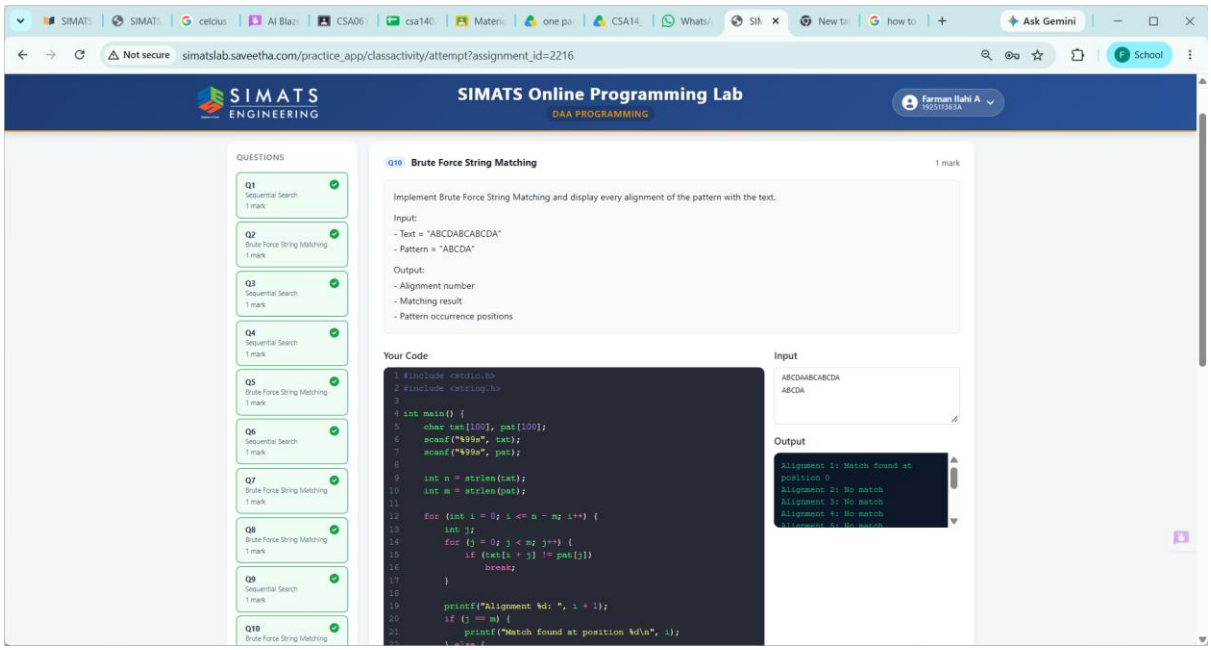
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QUESTIONS
Q8 Brute Force String Matching
1 mark
Implement Brute Force String Matching and display the comparison table for each alignment.
Input:
- Text = "MISSISSIPPI"
- Pattern = "ISSI"
Output:
- Shift number
- Comparisons made
- Match/Mismatch result
Your Code
Input
MISSISSIPPI
ISSI
Output
Shift 0: Comparisons = 1, Mismatch
Shift 1: Comparisons = 4, Match
Shift 2: Comparisons = 1, Mismatch

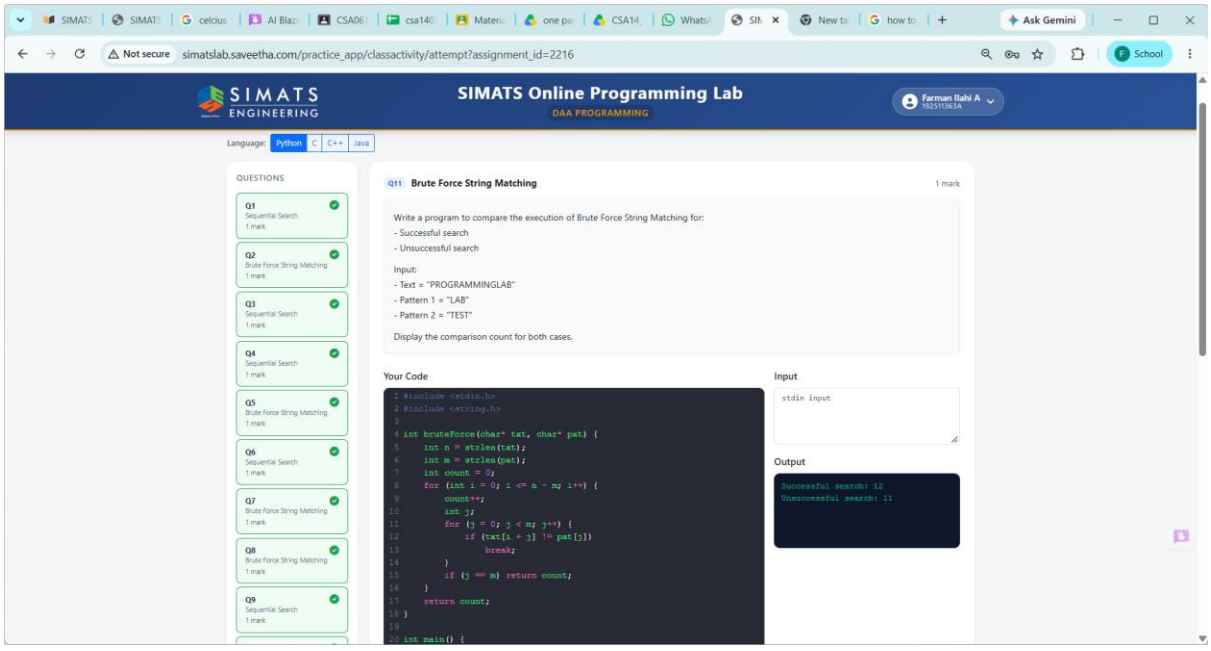
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QUESTIONS
Q9 Sequential Search
1 mark
Implement Sequential Search for an unsuccessful search.
Input:
Array: 5, 10, 15, 20, 25, 30, 35
Key: 18
Output:
Element not found
Number of comparisons = 7
Your Code
Input
5 10 15 20 25 30 35
18
Output
Element not found
Number of comparisons = 7

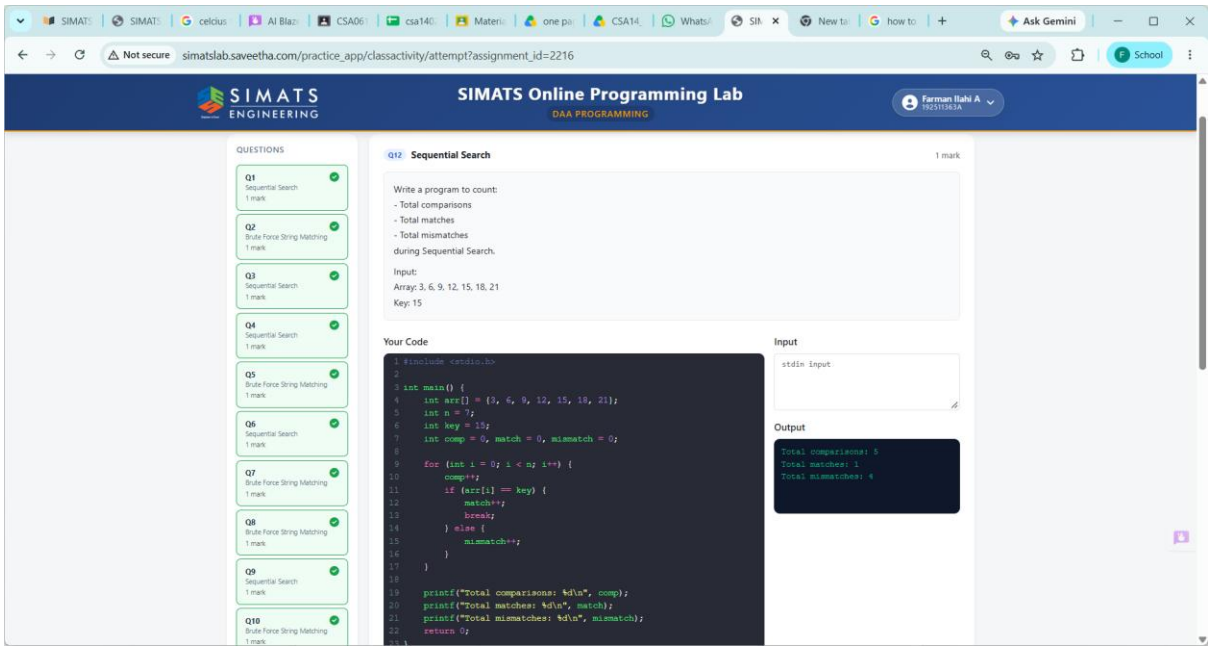
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QUESTIONS
Q10 Brute Force String Matching
1 mark
Implement Brute Force String Matching and display every alignment of the pattern with the text.
Input:
- Text = "ABCDABCABCDA"
- Pattern = "ABCDA"
Output:
- Alignment number
- Matching result
- Pattern occurrence positions
Your Code
Input
ABCDAABCABCDA
ABCDA
Output
Alignment 1: Match found at position 0
Alignment 2: No match
Alignment 3: No match
Alignment 4: No match

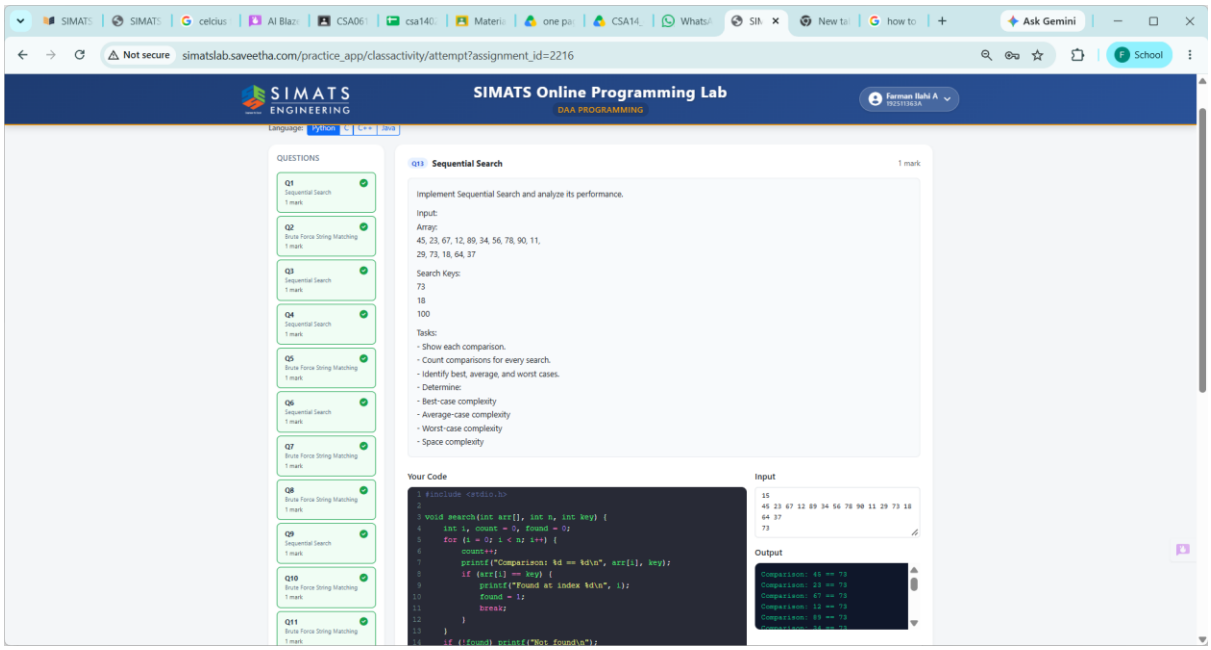
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Language: Python C C++ Java
QUESTIONS
Q11 Brute Force String Matching
1 mark
Write a program to compare the execution of Brute Force String Matching for:
- Successful search
- Unsuccessful search
Input:
- Text = "PROGRAMMINGLAB"
- Pattern 1 = "LAB"
- Pattern 2 = "TEST"
Display the comparison count for both cases.
Your Code
Input
stdin input
Output
Successful search: 12
Unsuccessful search: 11

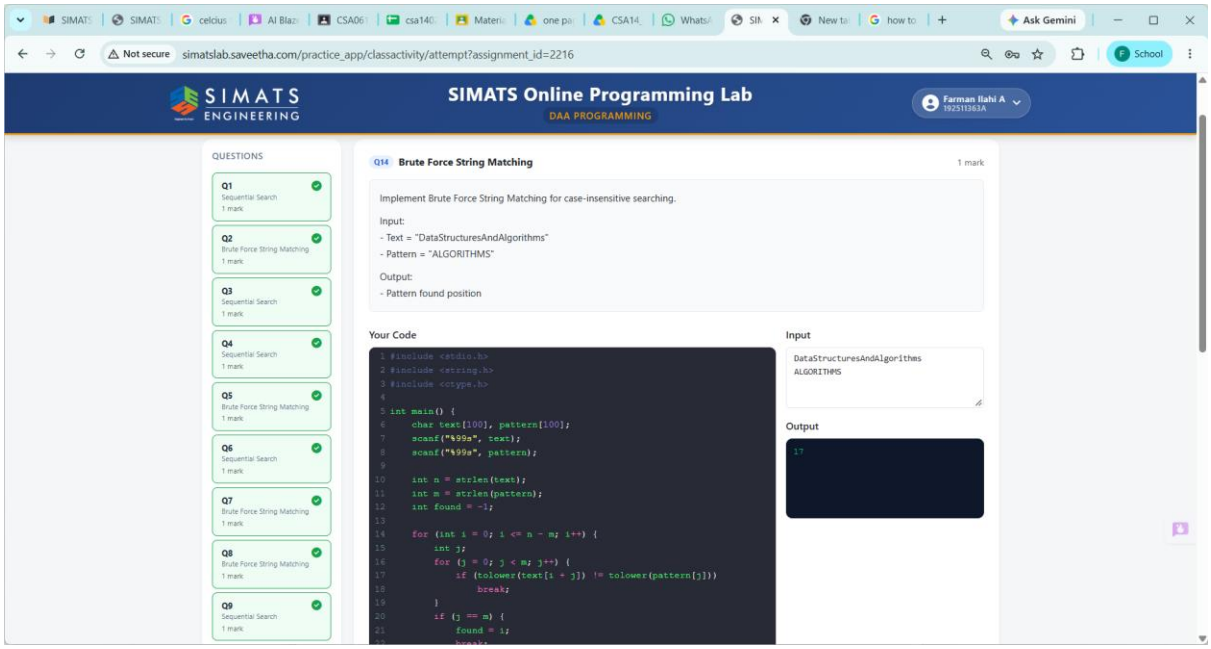
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QUESTIONS
Q12 Sequential Search
1 mark
Write a program to count:
- Total comparisons
- Total matches
- Total mismatches
during Sequential Search.
Input:
Array: 3, 6, 9, 12, 15, 18, 21
Key: 15
Your Code
Input
stdin input
Output
Total comparisons: 5
Total matches: 1
Total mismatches: 4

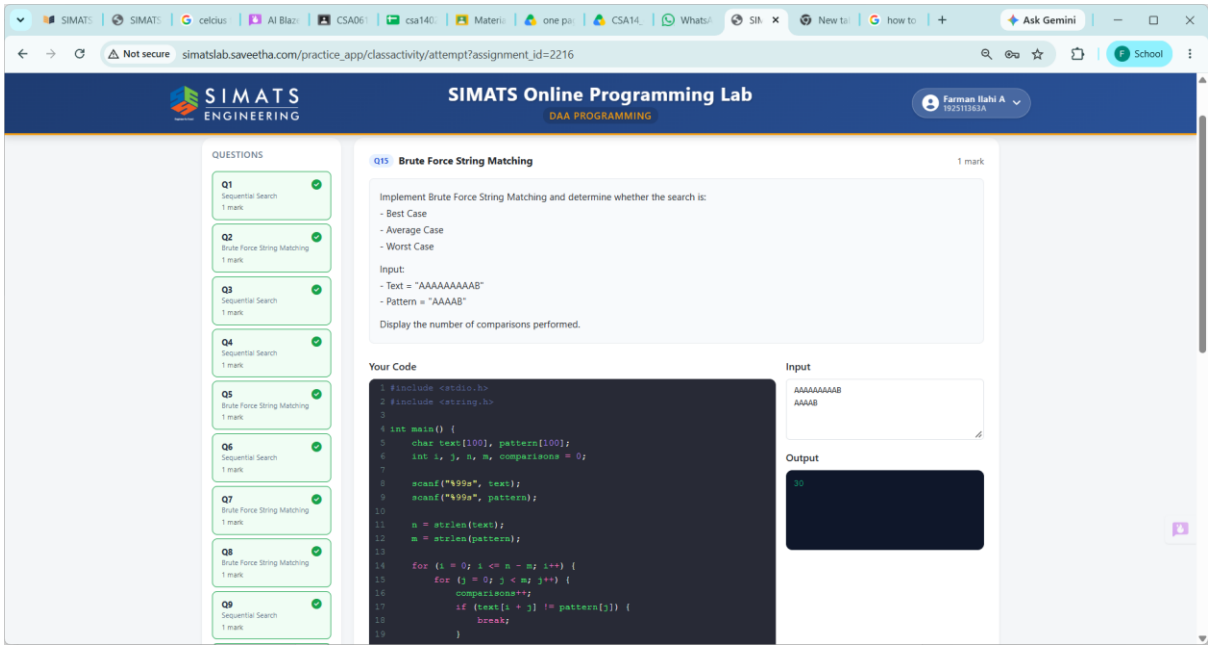
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QUESTIONS
Q13 Sequential Search
1 mark
Implement Sequential Search and analyze its performance.
Input:
Array:
45, 23, 67, 12, 89, 34, 56, 78, 90, 11,
29, 73, 18, 64, 37
Search Keys:
73
18
100
Tasks:
- Show each comparison.
- Count comparisons for every search.
- Identify best, average, and worst cases.
- Determine:
- Best-case complexity
- Average-case complexity
- Worst-case complexity
- Space complexity
Your Code
Input
Output

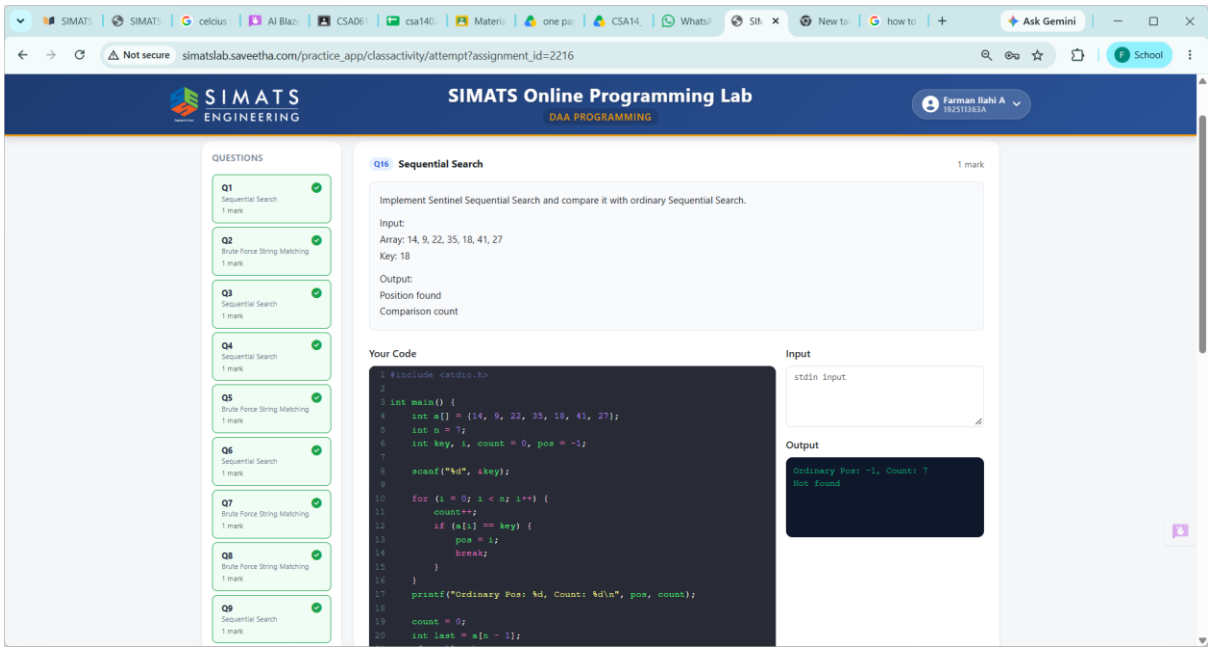
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QUESTIONS
Q1 Sequential Search 1 mark
Q2 Brute Force String Matching 1 mark
Q3 Sequential Search 1 mark
Q4 Sequential Search 1 mark
Q5 Brute Force String Matching 1 mark
Q6 Sequential Search 1 mark
Q7 Brute Force String Matching 1 mark
Q8 Brute Force String Matching 1 mark
Q9 Sequential Search 1 mark
Q14 Brute Force String Matching
1 mark
Implement Brute Force String Matching for case-insensitive searching.
Input:
- Text = "DataStructuresAndAlgorithms"
- Pattern = "ALGORITHMS"
Output:
- Pattern found position
Your Code
Input
DataStructuresAndAlgorithms
ALGORITHMS
Output
17

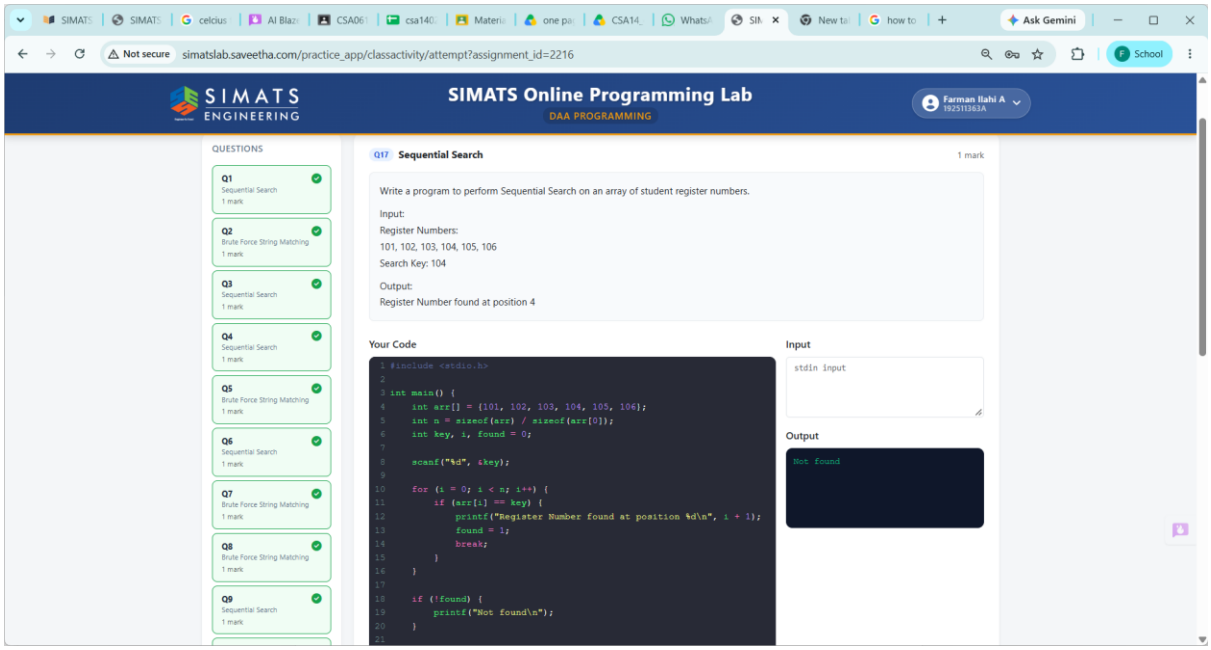
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QUESTIONS
Q1 Sequential Search 1 mark
Q2 Brute Force String Matching 1 mark
Q3 Sequential Search 1 mark
Q4 Sequential Search 1 mark
Q5 Brute Force String Matching 1 mark
Q6 Sequential Search 1 mark
Q7 Brute Force String Matching 1 mark
Q8 Brute Force String Matching 1 mark
Q9 Sequential Search 1 mark
Q15 Brute Force String Matching
1 mark
Implement Brute Force String Matching and determine whether the search is:
- Best Case
- Average Case
- Worst Case
Input:
- Text = "AAAAAAAAAB"
- Pattern = "AAAAB"
Display the number of comparisons performed.
Your Code
Input
AAAAAAAAB
AAAAB
Output
30

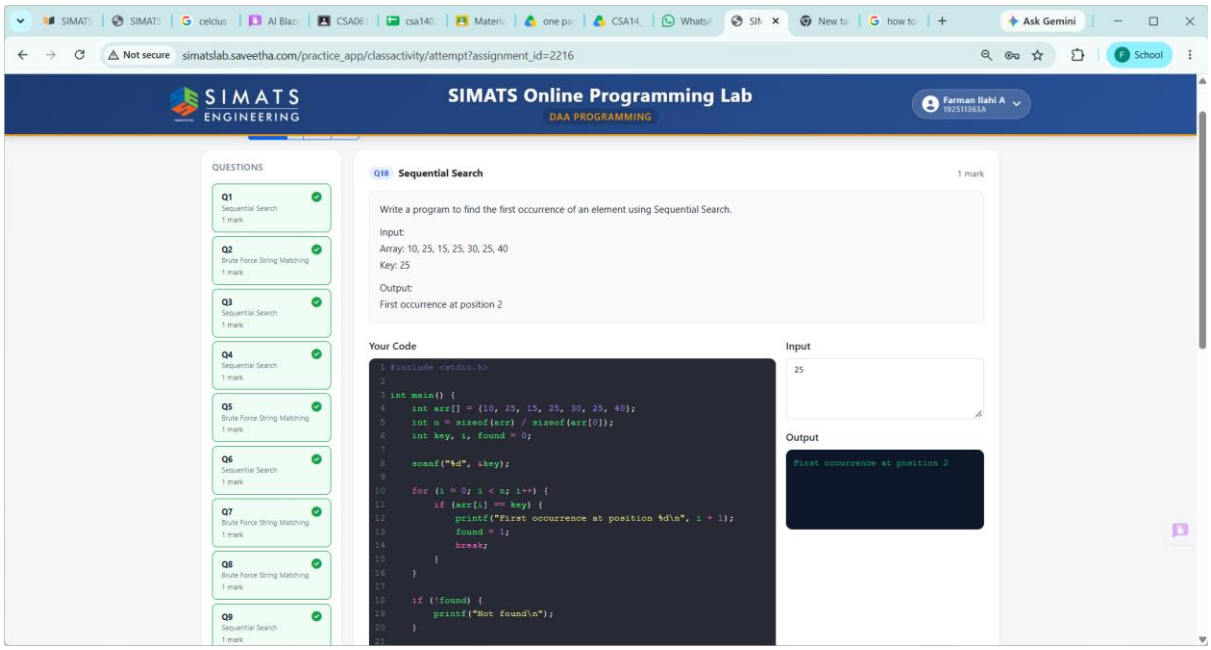
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QUESTIONS
Q16 Sequential Search
1 mark
Implement Sentinel Sequential Search and compare it with ordinary Sequential Search.
Input:
Array: 14, 9, 22, 35, 18, 41, 27
Key: 18
Output:
Position found
Comparison count
Your Code
Input
stdin input
Output
Ordinary Pos: -1, Count: 7
Not found

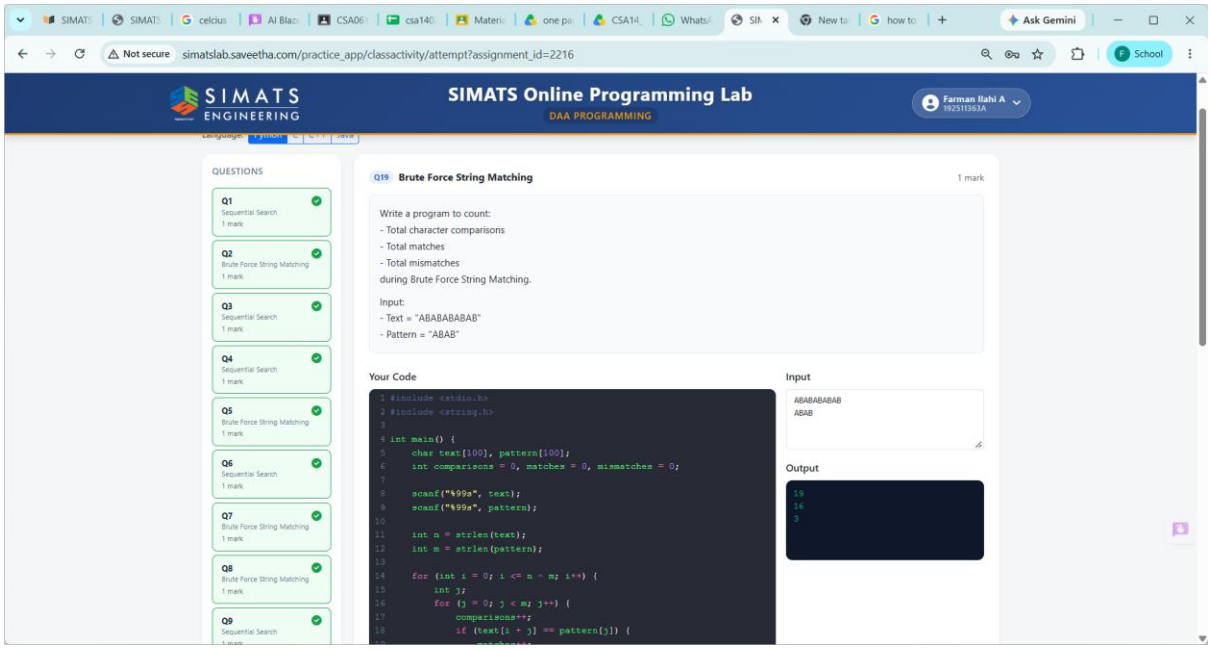
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QUESTIONS
Q17 Sequential Search
1 mark
Write a program to perform Sequential Search on an array of student register numbers.
Input:
Register Numbers:
101, 102, 103, 104, 105, 106
Search Key: 104
Output:
Register Number found at position 4
Your Code
Input
stdin input
Output
Not found

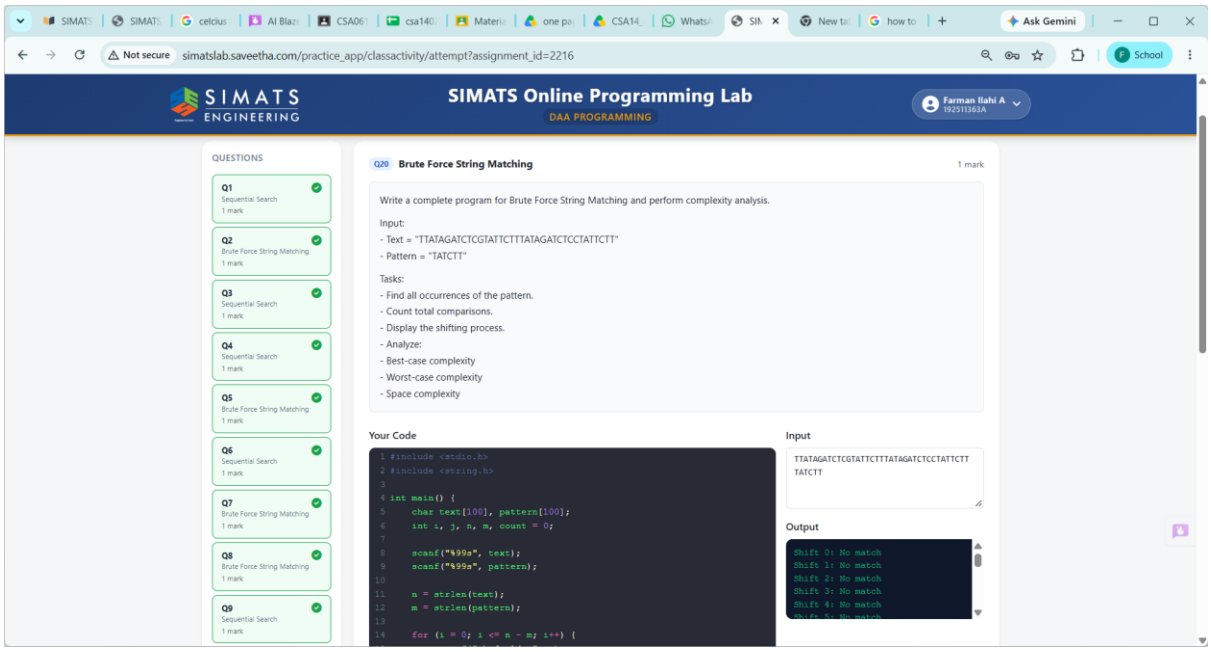
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QUESTIONS
Q1 Sequential Search 1 mark
Q2 Brute Force String Matching 1 mark
Q3 Sequential Search 1 mark
Q4 Sequential Search 1 mark
Q5 Brute Force String Matching 1 mark
Q6 Sequential Search 1 mark
Q7 Brute Force String Matching 1 mark
Q8 Brute Force String Matching 1 mark
Q9 Sequential Search 1 mark
Q10 Sequential Search
1 mark
Write a program to find the first occurrence of an element using Sequential Search.
Input:
Array: 10, 25, 15, 25, 30, 25, 40
Key: 25
Output:
First occurrence at position 2
Your Code
#include <stdio.h>
int main() {
int arr[] = {10, 25, 15, 25, 30, 25, 40};
int n = sizeof(arr) / sizeof(arr[0]);
int key, i, found = 0;
scanf("%d", &key);
for (i = 0; i < n; i++) {
if (arr[i] == key) {
printf("First occurrence at position %d\n", i + 1);
found = 1;
break;
}
}
if (!found) {
printf("Not found\n");
}
Input
25
Output
First occurrence at position 2

19)


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QUESTIONS
Q1 Sequential Search 1 mark
Q2 Brute Force String Matching 1 mark
Q3 Sequential Search 1 mark
Q4 Sequential Search 1 mark
Q5 Brute Force String Matching 1 mark
Q6 Sequential Search 1 mark
Q7 Brute Force String Matching 1 mark
Q8 Brute Force String Matching 1 mark
Q9 Sequential Search 1 mark
Q19 Brute Force String Matching
1 mark
Write a program to count:
- Total character comparisons
- Total matches
- Total mismatches
during Brute Force String Matching.
Input:
- Text = "ABABABABAB"
- Pattern = "ABAB"
Your Code
#include <stdio.h>
#include <string.h>
int main() {
char text[100], pattern[100];
int comparisons = 0, matches = 0, mismatches = 0;
scanf("%99s", text);
scanf("%99s", pattern);
int n = strlen(text);
int m = strlen(pattern);
for (int i = 0; i <= n - m; i++) {
int j;
for (j = 0; j < m; j++) {
comparisons++;
if (text[i + j] == pattern[j]) {
Input
ABABABABAB
ABAB
Output
19
16
3

20)


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QUESTIONS
Q1 Sequential Search 1 mark
Q2 Brute Force String Matching 1 mark
Q3 Sequential Search 1 mark
Q4 Sequential Search 1 mark
Q5 Brute Force String Matching 1 mark
Q6 Sequential Search 1 mark
Q7 Brute Force String Matching 1 mark
Q8 Brute Force String Matching 1 mark
Q9 Sequential Search 1 mark
Q20 Brute Force String Matching
1 mark
Write a complete program for Brute Force String Matching and perform complexity analysis.
Input:
- Text = "TTATAGATCTCGTATTCTTTATAGATCTCCTATTCTT"
- Pattern = "TATCTT"
Tasks:
- Find all occurrences of the pattern.
- Count total comparisons.
- Display the shifting process.
- Analyze:
- Best-case complexity
- Worst-case complexity
- Space complexity
Your Code
#include <stdio.h>
#include <string.h>
int main() {
char text[100], pattern[100];
int i, j, n, m, count = 0;
scanf("%99s", text);
scanf("%99s", pattern);
n = strlen(text);
m = strlen(pattern);
for (i = 0; i <= n - m; i++) {
Input
TTATAGATCTCGTATTCTTTATAGATCTCCTATTCTT
TATCTT
Output
Shift 0: No match
Shift 1: No match
Shift 2: No match
Shift 3: No match
Shift 4: No match